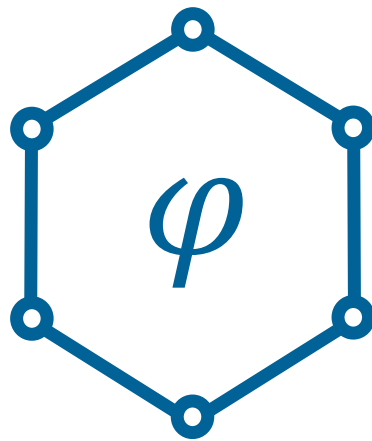




Logical Knowledge in KGs

Recursion in Datalog

Primer

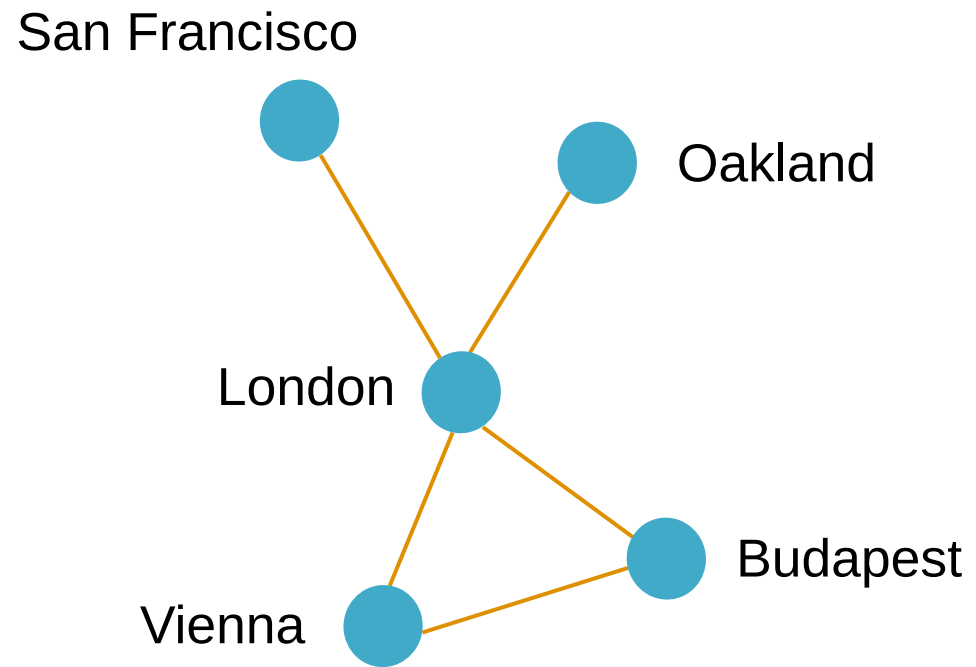
Emanuel Sallinger

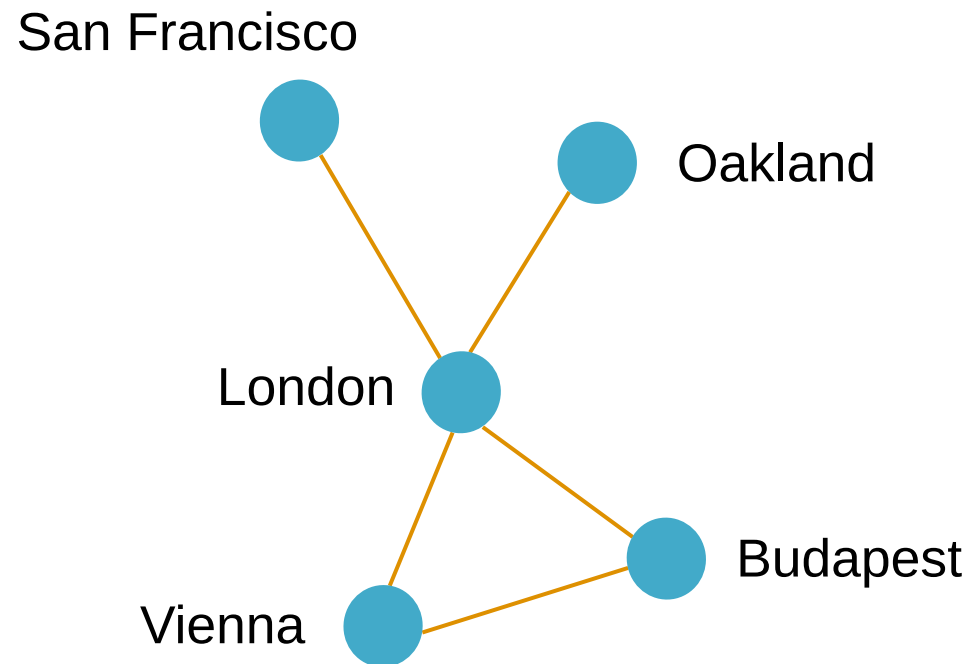
SQL

```
WITH RECURSIVE temp AS (  
    SELECT * FROM flight  
UNION ALL  
    SELECT t.Origin, f.Destination  
    FROM temp t, flight f  
    WHERE t.Destination = f.Origin  
)  
SELECT * INTO connection FROM temp
```

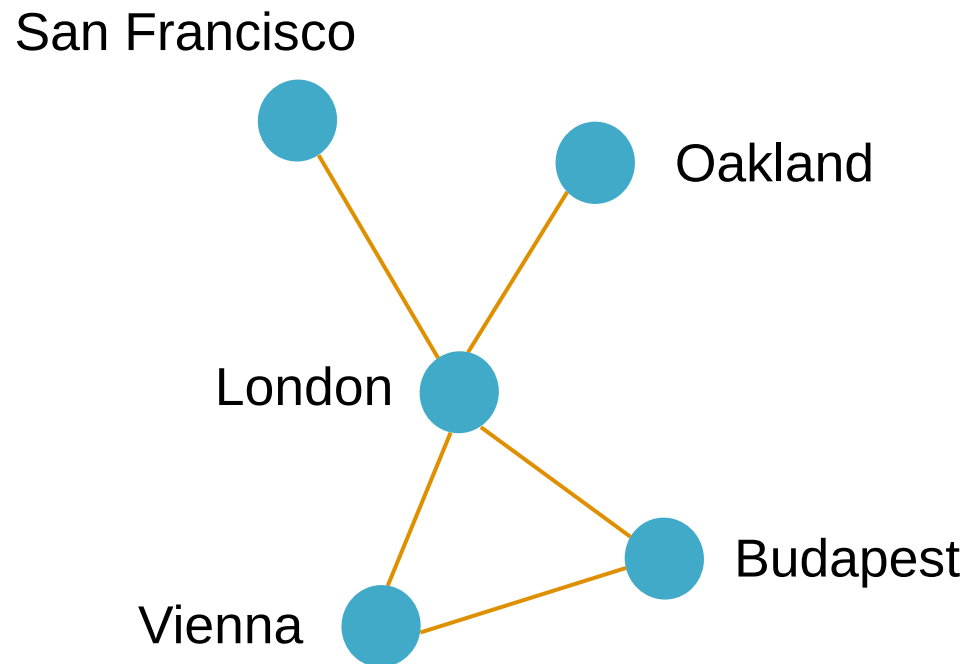
Datalog

```
connection(X,Y) :- flight(X,Y,_).  
connection(X,Z) :- connection(Y,Z), flight(X,Y,_).
```





airport	
IATA	City
VIE	Vienna
LHR	London
LGW	London
BUD	Budapest
OAK	Oakland
SFO	San Francisco



flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
VIE	Vienna
LHR	London
LGW	London
BUD	Budapest
OAK	Oakland
SFO	San Francisco

airport

IATA	City
VIE	Vienna
LHR	London
SFO	San Francisco



city

City
Vienna
London
San Francisco

airport

IATA	City
VIE	Vienna
LHR	London
SFO	San Francisco



city

City
Vienna
London
San Francisco

SQL

```
SELECT City INTO city FROM airport
```

airport

IATA	City
VIE	Vienna
LHR	London
SFO	San Francisco



city

City
Vienna
London
San Francisco

SQL

```
SELECT City INTO city FROM airport
```

Datalog

```
city(City) :- airport(IATA, City).
```



```
city(City) :- airport(IATA,City).
```

head

body



```
city(City) :- airport(IATA,City).
```

head

body



```
city(City) :- airport(IATA, City).
```

then

if

head

body



```
city(City) :- airport(IATA, City).
```

SELECT

FROM

```
city(City) :- airport(IATA,City).
```

```
city(City) :- airport(IATA,City).
```

table (or relation, edge, node, ML, etc.)

```
city(City) :- airport(IATA,City).
```

table (or relation, edge, node, ML, etc.)

```
SELECT City INTO city FROM airport
```

variable (not necessarily attribute name)

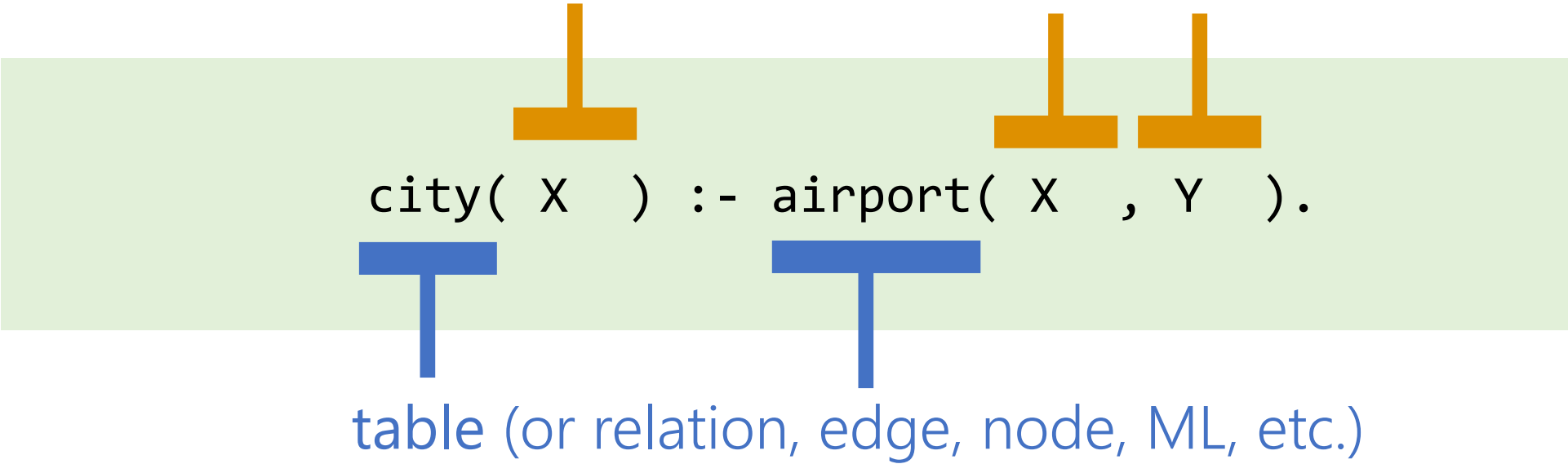


```
city(City) :- airport(IATA, City).
```

table (or relation, edge, node, ML, etc.)

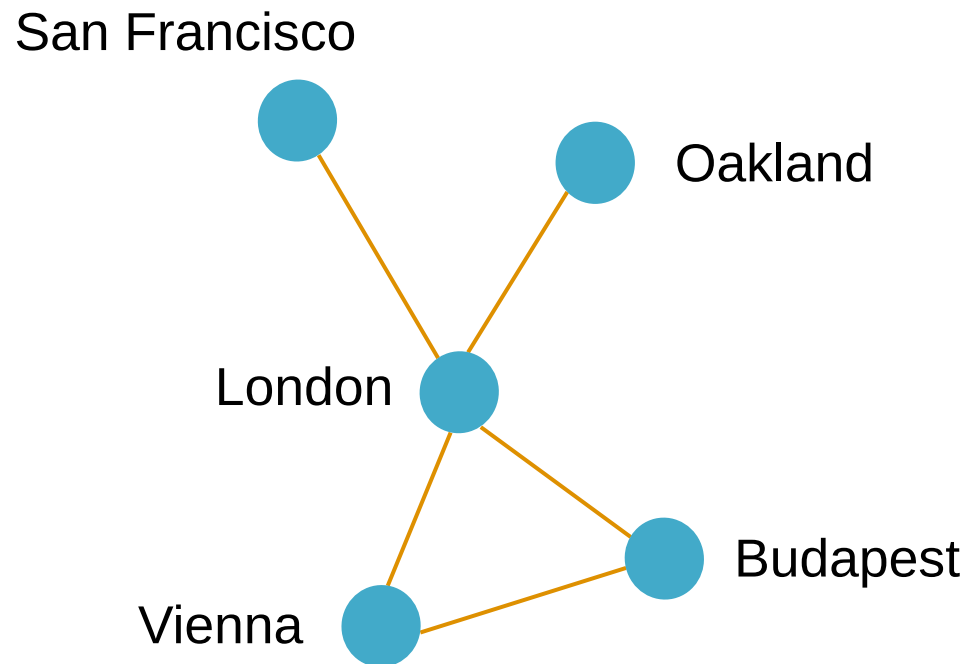
```
SELECT City INTO city FROM airport
```


variable (not necessarily attribute name)



```
city( X ) :- airport( X , Y ).
```

table (or relation, edge, node, ML, etc.)

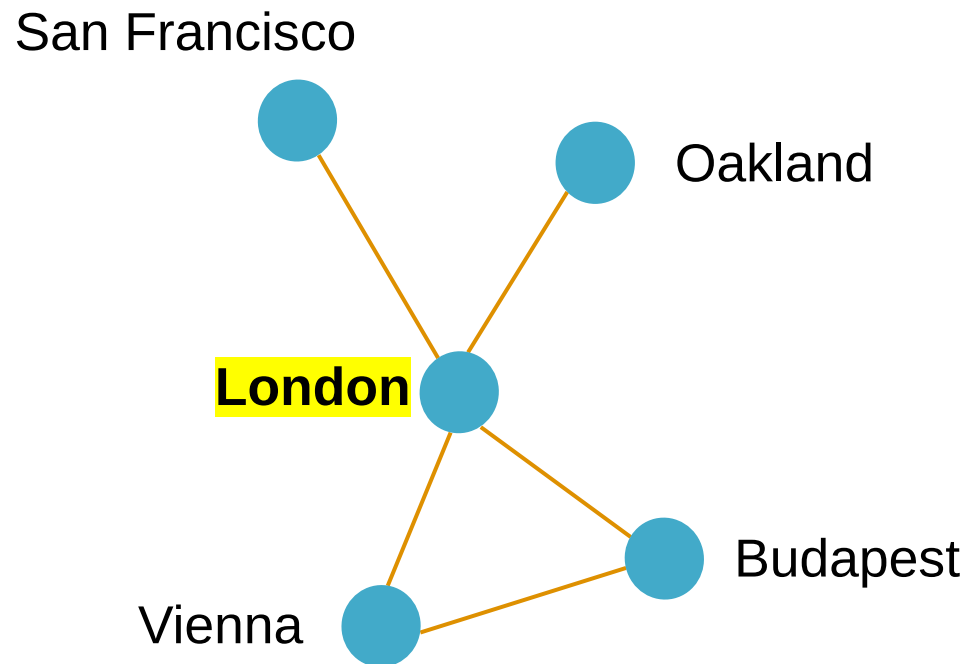


flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
VIE	Vienna
LHR	London
LGW	London
BUD	Budapest
OAK	Oakland
SFO	San Francisco

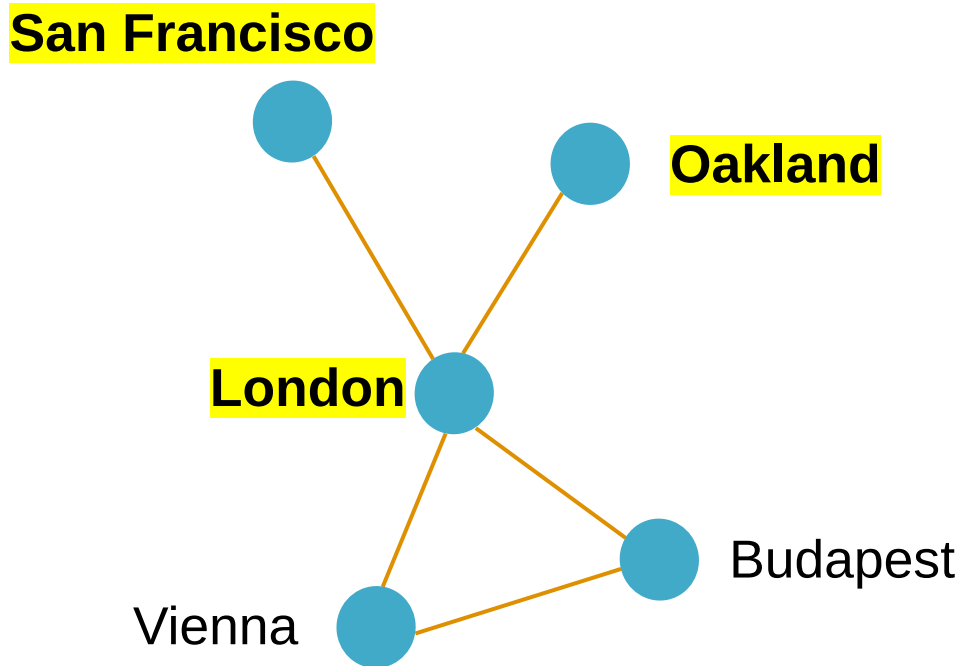


flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
VIE	Vienna
LHR	London
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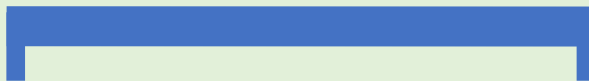
flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
VIE	Vienna
LHR	London
LGW	London
BUD	Budapest
OAK	Oakland
SFO	San Francisco

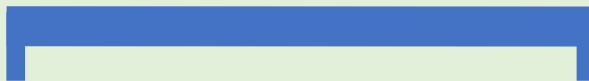
join



```
dest(D) :- flight(L,D,A), airport(L,"London").
```

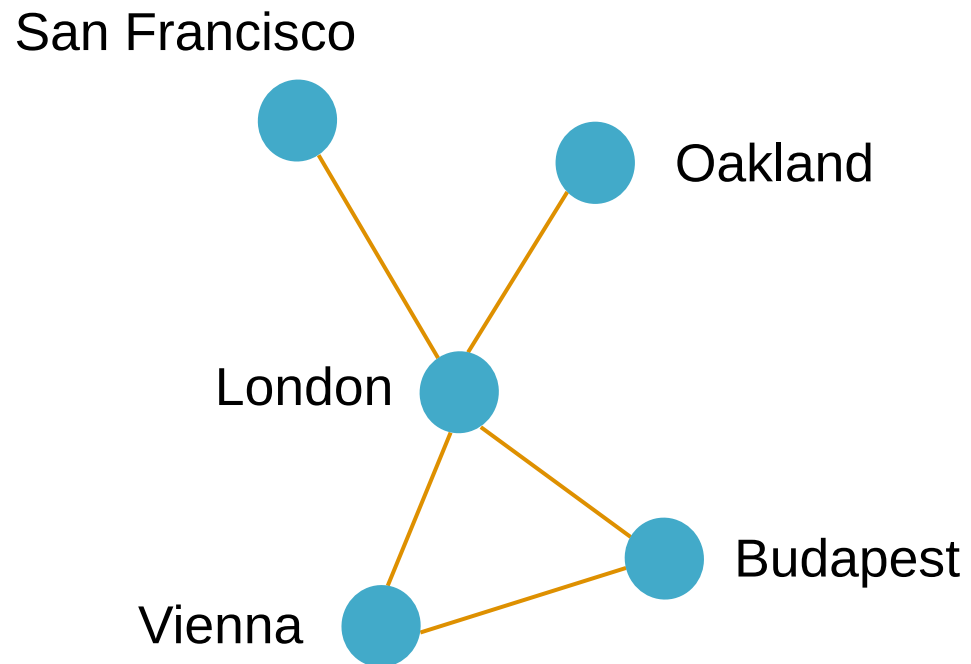
```
SELECT f.Destination INTO dest  
FROM flight f, airport a  
WHERE f.Origin = a.IATA AND a.City = "London"
```

join



```
dest(D) :- flight(L,D,A), airport(L,City),  
           City = "London".
```

```
SELECT f.Destination INTO dest  
FROM flight f, airport a  
WHERE f.Origin = a.IATA AND a.City = "London"
```

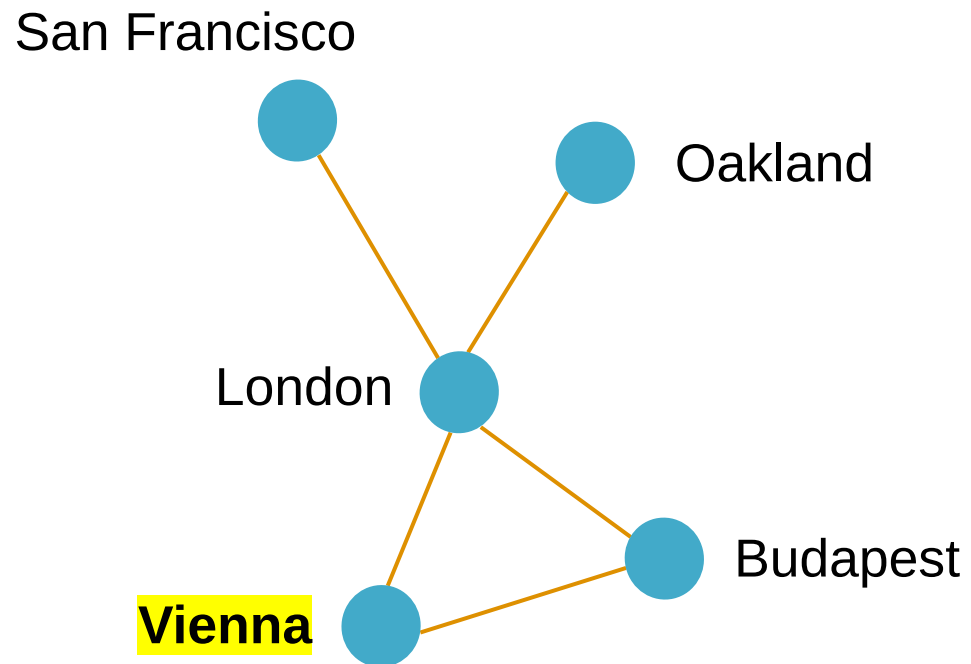


flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
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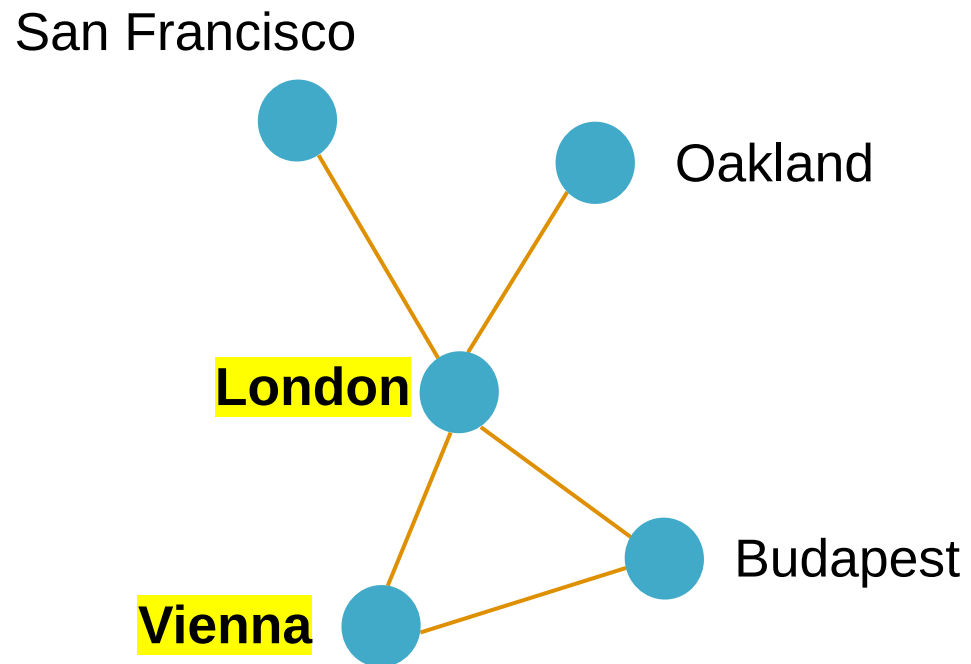


flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
VIE	Vienna
LHR	London
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BUD	Budapest
OAK	Oakland
SFO	San Francisco

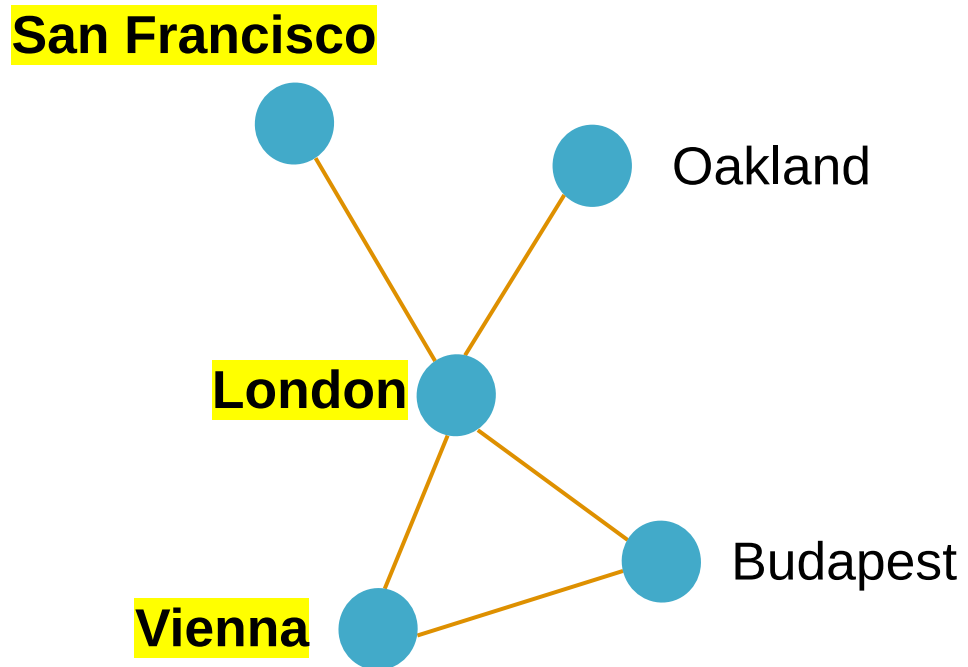


flight

Origin	Destination	Airline
VIE	LHR	BA
LHR	SFO	BA
LGW	OAK	DI
BUD	VIE	OS

airport

IATA	City
VIE	Vienna
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flight

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)  
SELECT * INTO connection FROM temp
```

Datalog

```
connection(X,Y) :- flight(X,Y,_).  
connection(X,Z) :- connection(Y,Z), flight(X,Y,_).
```

Syntax of Datalog

A **datalog rule** is an expression of the form

$$R_1(x_1) :- R_2(x_2), \dots, R_n(x_n).$$

where R_i are relation names and x_i are lists of variable names.

- Each variable in x_1 must occur in at least one of $x_2, \dots, x_n$.
- A **datalog programme** is a finite set of datalog rules.



See Abitebeoul, Hull and Vianu's
"Foundations of Databases" for more details

Semantics of Datalog

- Various semantics are possible:
 - Model-theoretic semantics
 - Fixpoint semantics
 - Proof-theoretic semantics
- Important tool: The “chase” procedure
 - Informally, starting with a database and applying rules to generate new tuples.
- More than we can cover here. For the interested:



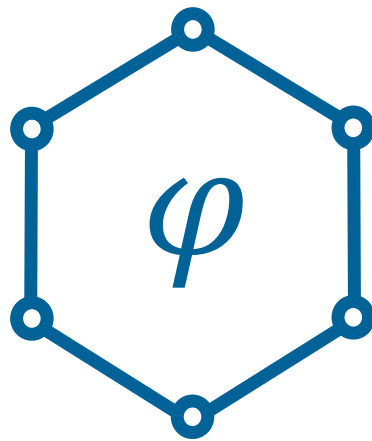
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Datalog

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connection(X,Y) :- flight(X,Y,_).  
connection(X,Z) :- connection(Y,Z), flight(X,Y,_).
```

First-order Logic

$$\forall X, Y, U \text{ (} \textit{connection}(X, Y) \leftarrow \textit{flight}(X, Y, U) \text{)}$$
$$\forall X, Y, Z, U \text{ } \textit{connection}(X, Z) \leftarrow \textit{connection}(Y, Z) \wedge \textit{flight}(X, Y, U)$$



Logical Knowledge in KGs

Recursion in Datalog

Primer

Emanuel Sallinger